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Sigmoid function, $\delta(z) = \frac{1}{1+e^{-z}}$

Derivative of this function

$$\delta'(z) = \frac{d}{dz} \left(\frac{1}{1+e^{-z}} \right)$$

$$= \frac{d}{dz} (1+e^{-z})^{-1}$$

$$= -\frac{1}{(1+e^{-z})^2} \cdot \frac{d}{dz} (1+e^{-z})$$

$$= -\frac{1}{(1+e^{-z})^2} \cdot -e^{-z}$$

$$= \frac{e^{-z}}{(1+e^{-z})^2}$$

$$= \frac{1}{1+e^{-z}} \cdot \frac{e^{-z}}{1+e^{-z}}$$

$$= \delta(z) \cdot \frac{e^{-z}}{1+e^{-z}} \left[\because \delta(z) = \frac{1}{1+e^{-z}} \right]$$

$$= \delta(z) \cdot \frac{1+e^{-z}-1}{1+e^{-z}}$$

$$= \delta(z) \cdot \left(\frac{1+e^{-z}}{1+e^{-z}} - \frac{1}{1+e^{-z}} \right)$$

$$= \delta(z) \cdot (1 - \delta(z))$$

$$\therefore \delta'(z) = \delta(z) \cdot (1 - \delta(z))$$